

Calibration-Aware Three-Dimensional Optical Positioning With a Fixed LED and a Tilt-Constrained Reorientable Photodiode

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Abstract—

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I. INTRODUCTION

A. Background and Motivation

B. Related Work and Research Gap

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II. SYSTEM AND MEASUREMENT MODEL

A. System Geometry

Consider a fixed LED at a known position $\mathbf{t} \in \mathbb{R}^3$, with unit optical axis $\mathbf{n}_t$ and constant transmitted power P_t . A single PD has an unknown position $\mathbf{r} = [x, y, z]^T$ and acquires RSS measurements at K orientations while its optical center remains stationary. The receiver normals $\mathbf{n}_i \in \mathbb{S}^2, i = 1, \dots, K$, are expressed in the same global coordinate frame and may be chosen arbitrarily. All three position coordinates are unknown.

The LED-PD distance and the unit direction from the PD toward the LED are:

$$d = \|\mathbf{t} - \mathbf{r}\|, \quad \mathbf{u} = \frac{\mathbf{t} - \mathbf{r}}{d}. \quad (1)$$

The irradiance angle ϕ and incidence angle ψ_i are determined by:

$$\cos \phi = -\mathbf{n}_t^T \mathbf{u}, \quad \cos \psi_i = \mathbf{n}_i^T \mathbf{u}. \quad (2)$$

Thus, receiver reorientation changes ψ_i , whereas d and ϕ remain constant during the acquisition sequence. The inclination of $\mathbf{n}_i$ relative to the global vertical is distinct from the incidence angle ψ_i ; no common inclination or azimuthal spacing is imposed.

B. Received-Power Model

Under line-of-sight (LOS) propagation and a Lambertian point-source model [1], [2], the calibrated received-power observation at orientation i is:

$$P_{r,i,k} = P_t H_i(\mathbf{r}) + \epsilon_{i,k}, \quad k = 1, \dots, N_i, \quad (3)$$

where N_i is the number of observations, $\epsilon_{i,k}$ is additive noise, and H_i is the effective LOS channel gain:

$$H_i(\mathbf{r}) = \frac{(m_T + 1)A_{PD}}{2\pi d^2} \cos^{m_T}(\phi) R(\psi_i). \quad (4)$$

Here A_{PD} denotes the active detector area. The LED Lambertian order is $m_T = -\ln 2 / \ln(\cos \Phi_{1/2})$, where $\Phi_{1/2}$ is its semi-angle at half power. Equation (4) applies for $0 \leq \phi < \pi/2$ and $0 \leq \psi_i < \Psi_{FOV} \leq \pi/2$; otherwise, $H_i = 0$. The receiver FOV is a half-angle. No optical concentrator is assumed, and the on-axis optical transmission is taken as unity. Received powers are expressed in optical-equivalent watts after background removal and gain/responsivity calibration.

C. Receiver Angular Response

The dimensionless function $R(\psi)$ describes the normalized effective angular response, with $R(0) = 1$. It includes the projected-area effect, so no additional factor $\cos \psi$ is required in (4). The receiver is assumed to have an axisymmetric response.

The conventional model uses $R(\psi) = \cos \psi$ [1], [2]. Experimental characterization of commercial PDs supports alternatives such as $R(\psi) = \cos^{m_R} \psi$, with $m_R > 0$, and the SQ model [3]:

$$R_{SQ}(\psi) = \left[1 - \frac{1 - 1/\sqrt{2}}{\psi_{3dB}^2} \psi^2 \right]_+, \quad (5)$$

where $[a]_+ = \max(a, 0)$ and ψ_{3dB} is the angle at which the response equals $1/\sqrt{2}$. The order m_R characterizes the receiver and is independent of m_T . The response model and its parameters are determined by calibration; the formulation retains $R(\psi)$ without selecting a particular family. Its response may vanish before the external FOV limit, whereas an angle of half response does not itself define a hard cutoff.

D. RSS Acquisition and Observation Statistics

At each orientation, observations are retained after the receiver has settled, as illustrated in Fig. 1. The baseline assumes linear nonsaturated detection and independent Gaussian optical-equivalent noise, $\epsilon_{i,k} \sim \mathcal{N}(0, \sigma_0^2)$, with calibrated variance σ_0^2 in W^2 . Defining $\mu_i(\mathbf{r}) = P_t H_i(\mathbf{r})$, the averaged observation satisfies:

$$\bar{P}_{r,i} = \frac{1}{N_i} \sum_{k=1}^{N_i} P_{r,i,k} \sim \mathcal{N}\left(\mu_i(\mathbf{r}), \frac{\sigma_0^2}{N_i}\right). \quad (6)$$

The observation vector $\bar{\mathbf{P}}_r = [\bar{P}_{r,1}, \dots, \bar{P}_{r,K}]^T$ therefore has covariance:

$$\Sigma_P = \text{diag}\left(\frac{\sigma_0^2}{N_1}, \dots, \frac{\sigma_0^2}{N_K}\right). \quad (7)$$

The total sample budget is $M = \sum_i N_i$; increasing K at fixed N_i and at fixed M are different comparisons. The per-observation and averaged-measurement SNRs are:

$$\text{SNR}_{i,\text{raw}} = \frac{\mu_i^2}{\sigma_0^2}, \quad \text{SNR}_{i,\text{mean}} = \frac{N_i \mu_i^2}{\sigma_0^2}. \quad (8)$$

The actual receiver normals are treated as known in this baseline; orientation uncertainty is introduced in Section IV. Noise correlation, calibration errors, and deviations from LOS propagation require separate characterization rather than being removed by averaging.

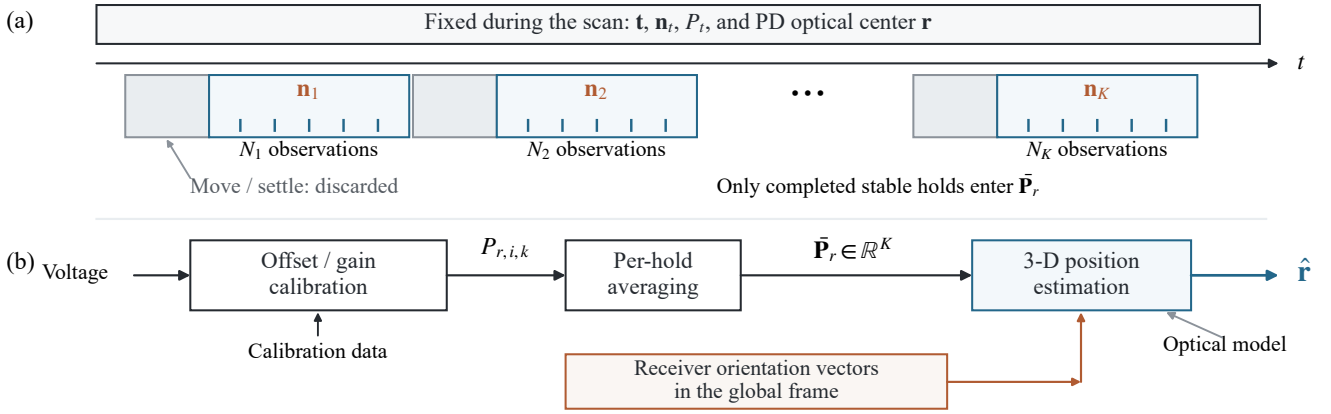


Fig. 1. Sequential RSS acquisition and processing. Transition intervals are discarded; N_i observations are retained at each stable orientation. Calibrated power means and globally referenced receiver normals form the inputs for position estimation.

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VIII. CONCLUSION

APPENDIX A

CHANNEL DERIVATIVES FOR THE CONSIDERED RESPONSE FAMILIES

APPENDIX B

ESTIMATION AND ORIENTATION-DESIGN PROCEDURES

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